

COLLABORATIVE FILTERING

LIKED BY ALICE AND BOB



SIMILAR USERS



LIKED BY ALICE, RECOMMENDED TO BOB

Collaborative Filtering & Hybrid Recommender Systems

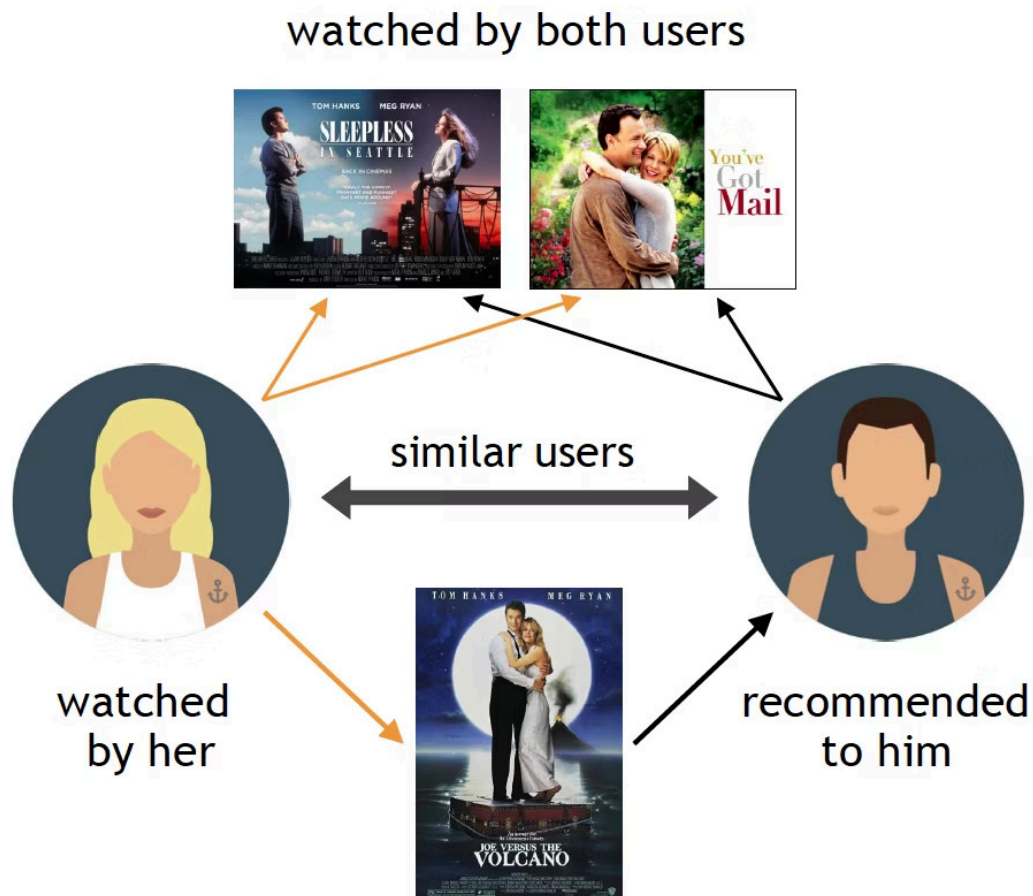
4/19/2026

Attendance - "Collaborative"



What is Collaborative Filtering?

Collaborative Filtering



Collaborative filtering recommends items by analyzing preference information from many users. It leverages the similarity of user preference behavior to make predictions.

Core principle: If some people have made similar decisions and purchases in the past, there is a high probability that they will agree on future selections.

This approach powers recommendations on platforms like Netflix, Amazon, and Spotify by identifying patterns across users.

How Collaborative Filtering Works

01

Construct User-Item Matrix

Build a matrix representing users and their ratings for each item in the catalog

02

Calculate Similarity

Compute similarity between users using cosine similarity or other distance metrics

03

Identify Similar Users

Find users with matching preference patterns and rating behaviors

04

Generate Recommendations

Recommend highly-rated items from similar users that have not been rated yet

	Item 1	Item 2	Item 3	Item 4	Item 5
User 1	0	3	0	3	4
User 2	4	0	0	2	0
User 3	0	0	3	0	0
User 4	3	0	4	0	3
User 5	4	3	0	4	4

Advantages of Collaborative Filtering



Discover New Interests

Helps users discover items that may not be entirely similar to previous items



No Contextual Data Needed

Metadata about items or users is not necessary, only the interactions between them



No Domain Knowledge Required

Does not need detailed item descriptions or domain expertise to function effectively

Challenges & Limitations

Cold Start Problem

Cannot handle new items because the model doesn't have training data for newly added items in the database

New User Difficulty

Struggles to recommend items to new users who lack rating history and preference data

Item Features Ignored

Does not leverage detailed item characteristics or content features in recommendations

Users	Movie 1	Movie 2	Movie 3	Movie 4
User 1	5	4		5
User 2	4		3	
User 3		1		2
User 4	1	2		

Hybrid Recommender Systems

Hybrid recommender systems combine content-based and collaborative filtering approaches to create more comprehensive, accurate recommendation engines.

1

Content-Based First

Initial recommendations use item features and descriptions when no user data exists

2

Collaborative Filtering Later

After system usage begins, collaborative methods leverage user interaction patterns



Netflix's Hybrid Approach

Onboarding Phase

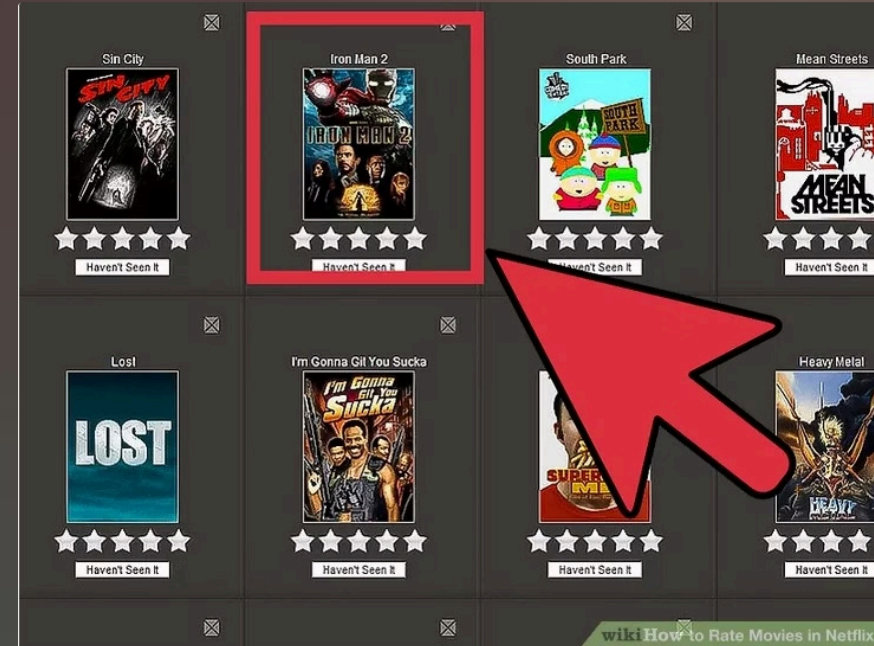
When a new user subscribes, Netflix requires them to rate content that they have already watched or indicate preferences for particular genres.

This initial data gathering enables content-based recommendations to start the personalization process.

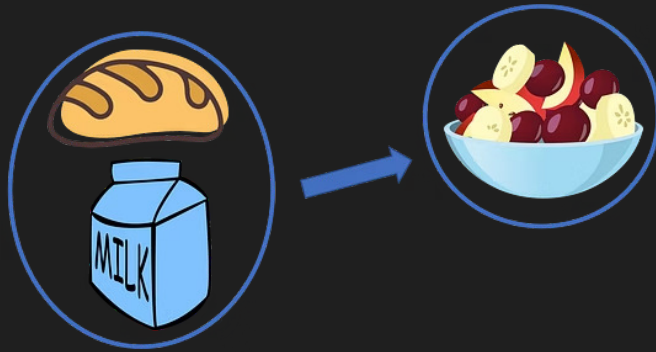
Active Usage Phase

Once the user begins engaging with the service, collaborative filtering is used.

The system identifies similar users and recommends content based on preferences and viewing patterns.



Association Rules Learning



Association Rule Learning

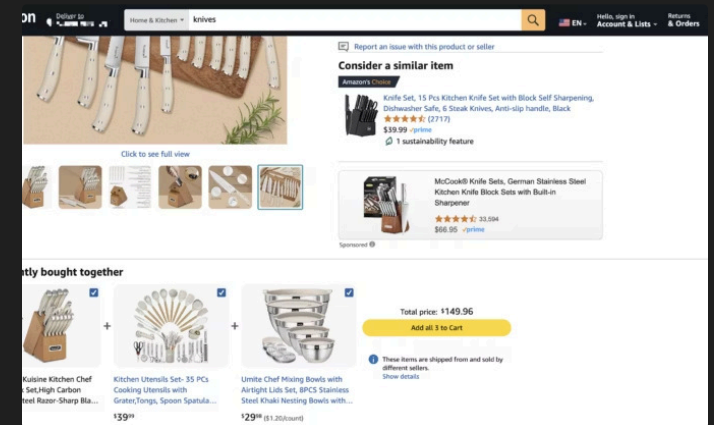
Hybrid Recommender Systems

Used to identify relationships between products



Retail Applications

Primarily used where users buy products paired with main items



Bundled Products

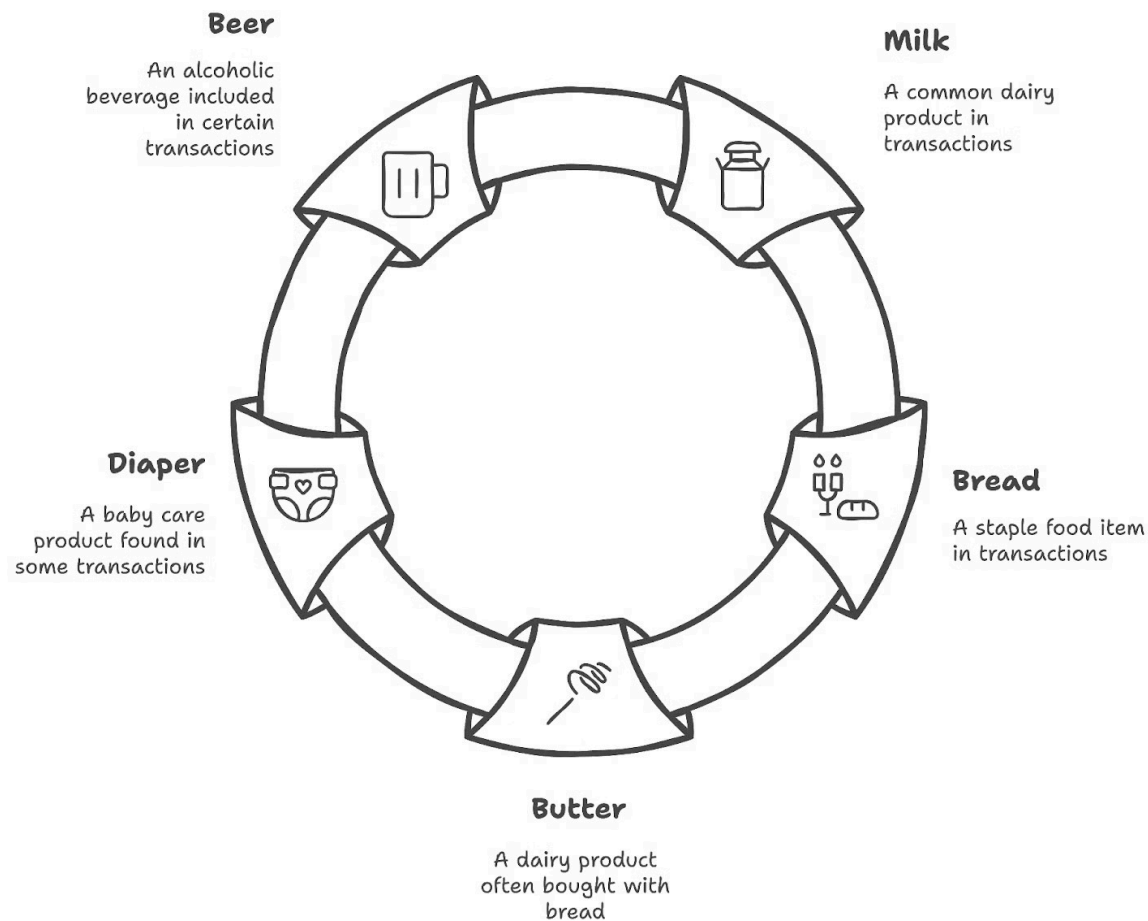
Recommends products that are commonly bought together

Google Colab



Apriori Algorithm Overview

Understanding Transaction Datasets



Purpose

Extracts frequent itemsets from transactional databases and generates association rules for recommendation systems.

Core Principle

If an itemset is frequent, all its subsets must also be frequent (downward-closure property)

Performance Note

Can be slow with large datasets since it generates many candidate itemsets during processing.

Apriori Algorithm Process



Generate Candidate Itemsets

Identify individual items and count occurrences to determine frequent items in transactions



Prune by Minimum Support

Remove itemsets that appear less than the minimum support threshold from consideration



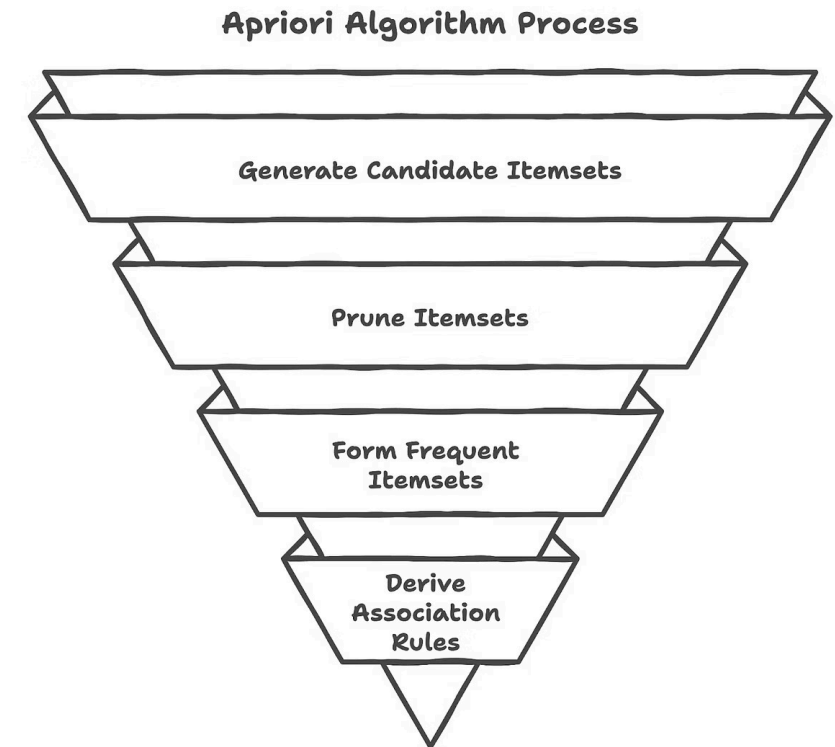
Build Frequent Itemsets

Generate larger itemsets by combining frequent smaller itemsets through iterative processes



Derive Association Rules

Extract meaningful rules based on confidence and lift values to determine item relationships



Thank you for coming!

Next Meeting: 4/26 - End-of-Year Social